Higher National Diploma in Information Technology

Second Year second Semester Examination-2021

HNDIT 2411 Enterprise Architecture

Question 1

1. Describe the term stateful and stateless in computing? (3 marks)
2. What is the Service Oriented Computing? State three differences between Object-Oriented Computing and Service-Oriented Computing. (5 marks)
3. State three elements of Service-Oriented Computing
4. Mention three differences between traditional software development and service-oriented computing. (3 marks)
5. Describe multi tired architecture and distributed multi tired architecture using two diagrams. (6 marks)

Question 2

1. What is concurrency in multiprogram? (2 marks)

2. State a difference between thread and a process. (2 marks)

3. Write a java program to create a thread using runnable interface. (5 marks)

4. State two differences between Statement and Prepared Statement. (4 marks)

5. Suppose you have been asked to read and view all the records of "Employee" table in the database called ABC Company. Employee table has three fields called empID, empName, and gender. Assume that the database is available in the localhost server. Fill in the blanks in the code segment given below to view all the data records of Employee table. (07 marks)

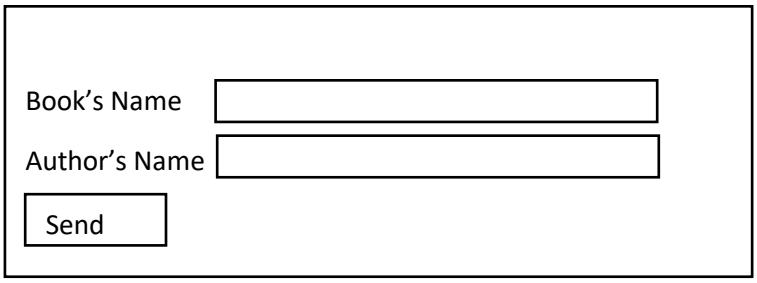
```
Public class DBConnectivty{
public static void main (String[] args){

        try{
        .....A..... ("com.mysql.jdbc.Driver");
        Connection con=Driver Manage....."jdbc:mysql://localhost: 3306/….c...." " root", ");
        Statement stmt=.....D....createStatement();
        ......E.. rs=stmt....F....("select * from Employee");
        while(rs.....G)
            System.out.printIn(rs.getString(1)+" "+rs.getString(2)+ " "+rs.getString(3));
        }
        }
        catch(Exception e){
            System.out.println(e);

        }

}

}
```

Question 3

1. What is the usage of deployment descriptor in java web application? (2 marks)

2. State two differences between general servlet and http servlet. (4 marks)

3. Suppose you have been asked to display book's name and author's name from the servlet called Display.java. The book's name and author's name are displaying which are given by the users from the html page called Data.html. Following image is a fragment of Data.html. Fill in the blanks in Data.html and Display.java. (8 marks)



Data.html

```
<html>
  <head>
    <title>Form data</title>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
   </head>
   <body>
      <form method=".....A........" action=".....B.....">
          First Name<input type="text" name="bvalue"><p>
          Last Name<input type="text" name="avalue"><p>
          <input type="submit" value="Send">
      </form>
 </body>
</html>
```

Display.java

```
        public class Display extends .....C.... {
        protected void process Request.......D..... request,.....E..... response) throws
        ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = .....F.....getWriter()) {
            String bookName=......G.....("bvalue");
            String authorName=......H....."avalue");
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet Display</title>");
            out.printin("</head>");
            out.printin("<body>");
            out, println'" <h1>Book detials " + bookName +""+ authorName + "</h1)");
            out.println("</body>");
            out.printin("</html>");

    }
}
```

4. Mention three session handling techniques in J2EE.

5. Explain the java code segment given below.

```
String getvalue=request.getParameter("userName");

out.print("Welcome "+getvalue);

out.print("<a href='servlet2?uname="+getvalue+"'>visit</a>");
```

Question 4

1. State two differences between HTML and XML. (03 marks)

2. Mention three attributes of syntactically well-formed XML document. (03 marks)

3. what is the use of DTD in XML. (03 marks)

4. what is XML parcel? (03 marks)

5. Write DTD for the XML file shown below. (08 marks)

```
</xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE note SYSTEM "doc.dtd">
<mail>
<to>Test1</to>
<from>Test2</from>
<subject>Contact</subject>
<Message>pleace contact Mr.ABC </message>
<date>27th may</date>
<time>5.00 p.m</time>
</mail>
```

Question 05

1. state two difference between scriplet and declaration elements in JSP. (04 marks)

2. write jsp code segment to perform the tasks shown below using correct element types in jsp.

   a. current date and time
   b. find the maximum number between two numbers. (08 marks)

3. what would be the output of the jsp code segment given below. (04 marks)

```
<html>
<body>
<%!
Int find(int x) {
return x*x*x;
}
%>
<%="The output is:"+find(4)%>
</body>
</html>
```

4. write the usage of the jsp standard actions shown bellow. (04 marks)

   a. <jsp:include>
   b. <jsp:forward>

Question 06

1. state why use SOAP. (02 marks)

2. Describe SOAP structure. (04 marks)

3.what is WSDL? What is the use of WSDL port. (04 marks)

4. name and describe two type of enterprise java beans. (04 marks)

5. Explain MVC architecture. (06 marks)